“BRIDGING SILENCE” : REAL TIME SIGN LANGUAGE TRANSLATION WITH WEBCAM

STEM PROJECT REPORT

SUBMITTED BY

SOWMITHA P U	710724121055
PAVITHRA R	710724121040
MOHANA MITHRA S	710724121032
SUPRIYA N	710724121058

BACHELOR OF ENGINEERING

IN

BIOMEDICAL ENGINEERING

Dr. N.G.P. INSTITUTE OF TECHNOLOGY, COIMBATORE

(An Autonomous Institution)

BONAFIDE CERTIFICATE

Certified that this project report "BRIDGING SILENCE: REAL TIME SIGN LANGUAGE TRANSLATION WITH WEBCAM" is the bonafide work of "SOWMITHA P U(710724121055), PAVITHRA R(710724121040), MOHANA MITHRA S(710724121032) & SUPRIYA N(710724121058) " who carried out the project work under my supervision**.**

SIGNATURE **SIGNATURE**

HEAD OF THE DEPARTMENT
Dr.S.Praveen Chakkravarthy
Professor and Head,
Department of Biomedical Engineering
Dr.N.G.P. Institute of Technology,
Coimbatore-641048

SUPERVISOR
Dr.J.Revathi
Asst.Prof(SG)/
Department of Biomedical Engineering,
Dr.N.G.P. Institute of Technology,
Coimbatore-641048

Submitted for the Project and Viva voce held on……..…..

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

We are highly thankful to our beloved Chairman Dr. Nalla G. Palaniswami M.D., A.B. (USA), and our respected Secretary, Dr. Thavamani D. Palaniswami M.D., A.B. (USA), FAAP who encouraged us throughout our career giving support and advice for shaping us to this level.

We greatly thank our Principal, Dr. S. U. Prabha M.E., Ph.D., for helping us to do our project successfully.

We heartily thank Dr.S.Praveen Chakkravarthy Head of Department Biomedical Engineering for his anchoring support in doing this project.

We are grateful to Project Coordinator, Dr.J.Revathi Assistant Professor, Department of Biomedical Engineering and for her advice and encouragement throughout the project.

We express our deepest gratitude to our guide Dr.J.Revathi Assistant Professor, Department of Biomedical Engineering for her unstrained support and technical assistance.

We also express our thanks to all the teaching and non-teaching staff members in the Department of Biomedical Engineering for their encouragement throughout the period of the project. Above all we thank the Lord Almighty for all the blessings and our endeavors

SOWMITHA P U
PAVITHRA R
MOHANA MITHRA S
SUPRIYA N

ABSTRACT

A real-time sign-to-speech system is a revolutionary technology that allows mute people to speak directly to their audience. Through the power of AI-powered sign recognition, natural language processing, and expressive text-to-speech technology, this system facilitates smooth communication. With the capability to record sign language in real-time through a regular webcam, it interprets signs into text and then translates it into natural-sounding speech. This process makes sure that deaf people can communicate naturally and smoothly, without the requirement of interpreters or captioning.

The sophisticated visual recognition capability of the system monitors hand movement, facial expression, and body movement, which are the core components of sign language. The system can use this technology to capture sign language accurately and with precision, which helps the system to grasp the subtlety of sign language and convert it into text. LLaVA NLP then comes into action, recognizing whole sentences, context, and feeling to give correct translations. This allows the system to recognize the nuances of sign language and transmit the intended meaning successfully.

Lastly, Kokoro AI translates the interpreted voice into clear, expressive, and contextually suitable speech. This is done to ensure that the voice produced is not just intelligible but also interesting, incorporating tone, rhythm, and pauses that a human would use naturally. The outcome is a natural voice that captures the message well, allowing listeners to easily understand and interact with the presentation.

By providing mute presenters with a voice, this system eliminates barriers and encourages interaction. Listeners are able to relate better to speech than silent or text presentations, and so presentations become more interactive and participative. From academic debates to boardroom meetings, or conferences, mute people can now be on an equal footing, conveying their thoughts without interruptions or need for external support. This can result in improved communication, more participation, and a more open environment.

The reach of this technology goes beyond the direct advantages of enhanced communication. By encouraging autonomy and equal contribution for the mute, it has the potential to make a powerful difference in almost every area of life, from education and the workplace to social interaction and beyond. With this system, mute individuals can communicate freely and assertively, unaided by interpreters or captioning systems. This can result in enhanced opportunities, greater self-esteem, and increased feelings of empowerment.

Finally, the real-time sign-to-speech system can transform communication for mute people. Through the use of AI-based technologies, it offers a natural and smooth means for mute people to interact with others. As the technology advances, it is bound to have an even greater impact on the lives of mute people, allowing them to engage fully in all spheres of society.

TABLE OF CONTENTS

CHAPTER NO:	TITLE	PAGE NO
1	**INTRODUCTION**	1
	1.1 Sign language translator	1
	1.2 Existing System	1
	1.3 Disadvantages	2
	1.4 Proposed System	2
2	**LITERATURE SURVEY**	3
	2.1 Introduction	3
	2.2 Literature Survey on sign language translator using webcam	
3	**METHODOLOGY**	5
	3.1 Introduction	5
	3.2 Block diagram	5
4	**COMPONENTS**	9
	4.1 Hardware components	9
	4.1.1 webcam	9
	4.2 Software	13
	4.2.1 LLAVA-NLP	13
	4.2.2 KOKORO-AI	14
5	**RESULTS AND DISCUSSION**	16
6	**CONCLUSION**	18
	REFERENCES	19

LIST OF FIGURES

FIGURE NO	TITLE	PAGE NO
3.1	Block Diagram	5
3.2	Circuit Diagram	6
4.1	Node MCU (ESP8266)	9
4.2	Motor Driver (L298)	10
4.3	Gear Motor	11
4.4	Temperature Sensor (MLX90614)	12
4.5	Arduino Software	13
4.6	Wifi Controller	14
4.7	ThingsBoard Software	15

CHAPTER-1

INTRODUCTION

1.1 SIGN LANGUAGE TRANSLATOR

Webcam sign language translators are revolutionary devices meant to give voiceless individuals the power to present and conduct seminars fearlessly. The technology uses machine learning and artificial intelligence to decode sign language hand signals from standard webcams and translate them into speech or text in real-time.

The effects of using such webcam-based sign language translators are significant. They allow mute presenters to present their ideas in an effective manner without relying on human interpreters, creating independence and inclusiveness.

By translating in real-time, these technologies eliminate communication barriers, enabling mute individuals to fully participate in seminars and discussions. This not only increases their contribution but also enhances a more inclusive atmosphere in schools and workplaces.

1.2 EXISTING SYSTEM

One of the prominent instances of a webcam-based sign language translation model is the Real-time Singapore Sign Language Translation System implemented by researchers through computer vision and Long Short-Term Memory (LSTM) networks.

The system records sign language gestures through a regular webcam, analyzes the visual data to identify significant features, and converts them into text in real time. The application of LSTM networks enables the model to address the temporal dependencies inherent in sign language, thereby enhancing dynamic gesture.

1.3 DISADVANTAGES

Singapore Sign Language Translator is a significant step forward, it has limitations such as reliance on specialized sensors and potential issues with gesture similarity and background noise.

It lacks standardization, locally developed signs, which can create inconsistencies. SgSL relies heavily on context, including facial expressions, body language, and spatial referencing, making it challenging for automated translation systems to capture accurately.

SgSL also captures unique cultural expressions that may be difficult to accurately convey in other languages. Moreover, there is a shortage of professionally trained SgSL interpreters, limiting access in critical areas like education, healthcare, and legal services.

1.4 PROPOSED SYSTEM

To address these challenges, we propose an innovative real-time sign language translation system that integrates advanced technologies—LLaVA-NLP and Kokoro TTS—to facilitate seamless and autonomous communication.

Allows mute individuals to participate without needing a human interpreter, promoting self-reliance.Provides immediate translation, ensuring seamless communication without delays.

Its innovative design ensures rapid processing and clear text and audio output, making it well-suited for real-time application. Its ability to provide accurate, expressive, and real-time translations using standard webcams makes it a valuable tool for empowering mute individuals to participate independently.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

For AI-powered sign language translation to work in everyday life, it must be both lightning-fast and natural-sounding. Conversations should flow effortlessly, without delays or robotic speech disrupting the interaction. That's where this game-changing technology comes in.

This groundbreaking AI system harnesses the latest multimodal models—"LLaVANLP" for sign language interpretation and "Kokoro TTS" for speech synthesis—to provide real-time, accurate, and expressive translation.

2.2 LITERATURE SURVEY ON SIGN-LANGUAGE TRANSLATORS

[1]. Advancing human-computer interaction(**2024**): this project uses convolutional neural networks (CNNs) to translate American Sign Language (ASL) into Nepali text and speech. It accurately recognizes ASL gestures from real-time video, converting them into Nepali text and speech through a text-to-speech engine.

[2].-CSLNSpeech(**2024):**it integrates Chinese Sign Language (CSL) alongside audio and facial data for improved speech separation. Using the CSLNSpeech dataset, the model significantly enhances separation accuracy, even in challenging conditions, demonstrating CSL's potential for aiding communication and learning for the hearing-impaired.

[2]. Building a largest Saudi sign language dataset(**2023):** This study develops the largest Saudi Sign Language (SSL) dataset, capturing diverse gestures through videos and images. It aims to support machine learning systems for automatic SSL recognition, emphasizing the need for culturally specific datasets in Arabic sign language research.

[4].American Sign Language to low-resource languages in Nigeria from Scientific African(**2023**): translating American Sign Language (ASL) into low-resource Nigerian languages like Hausa, Yoruba, and Igbo. It uses computer vision and natural language processing to interpret ASL gestures, addressing the lack of digital linguistic resources.

[5]. Voice Disorder Classification Using Speech Enhancement and Deep Learning Models(**2022**): combines speech enhancement and deep learning to improve voice disorder diagnosis. By integrating denoising techniques like spectral subtraction with CNNs and RNNs, the approach enhances signal quality, boosting classification accuracy, particularly in noisy environments.

[6]. De-centering and Negotiating Expertise(**2022**): Youth Voices and Language Play in Academic Discourses" examines how young people challenge traditional academic norms by integrating their voices, identities, and creative language use.

[7].Vision-based Hand Gesture Recognition Using Deep Learning for the Interpretation of Sign Language(**2021**): uses deep learning, specifically convolutional neural networks (CNNs), to accurately interpret sign language gestures from images or videos. This approach enables real-time translation into text or speech, enhancing accessibility for the hearing-impaired and promoting inclusive human-computer interaction.

CHAPTER 3

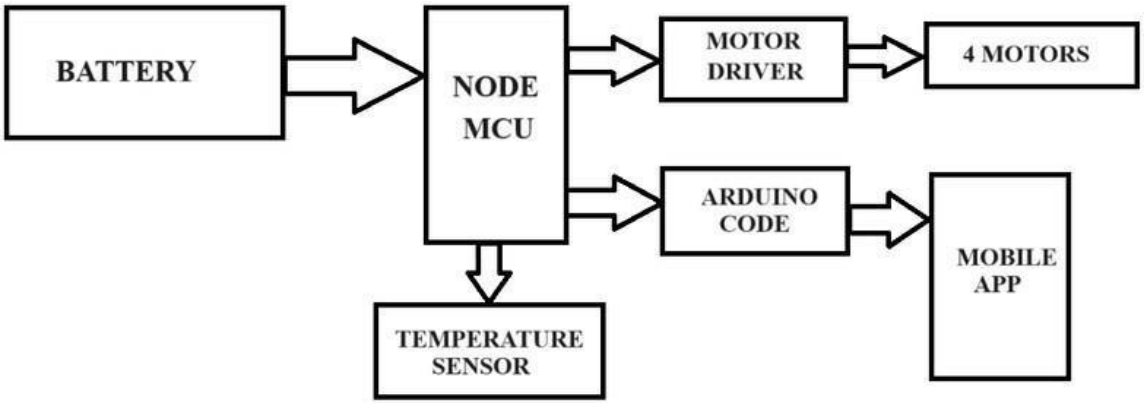
METHODOLOGY

3.1 INTRODUCTION

This approach outlines an end-to-end system for translating sign language into verbal language with the help of LLAVA- NLP and KOKORO-AI. The initial step involves the recording of the video via a webcam, which is the key input for sign language recognition. These frames or images are subsequently processed by Lava NLP, a robust natural language processing framework that identifies key features from the recorded visual data. Lava NLP translates these signs into the relevant text . The process is important as it transforms raw visual inputs into systematic text data that can be read and processed by language models easily.

After the text is generated, it is sent to KOKORO-AI that renders this text into natural, human-style speech. The last stage enables the interpreted sign language to be delivered as verbal language, with the effect of creating an uninterrupted and interactive communication for both hearing and speech-impaired. This combined process not only increases accessibility but also facilitates more inclusive human-computer interactions and fills the communication gap between the hearing and deaf population.

3.2 BLOCK DIAGRAM



The system begins by capturing real-time images or video frames of the signer using a webcam.This step is crucial as it provides the raw visual input required for recognizing and interpreting sign language.High-resolution cameras are preferred to capture the finer details of hand movements and facial expressions that are essential for accurate gesture recognition.

The captured frames are then processed using Lava NLP, a sophisticated natural language processing framework. Lava NLP first applies image preprocessing techniques, such as background removal, noise reduction, and hand segmentation, to isolate the hand gestures.These preprocessed frames are analyzed using deep learning models like convolutional neural networks (CNNs) to detect and classify the gestures.

Once recognized, the gesture data is converted into corresponding text, ensuring that the output maintains proper grammar and sentence structure.This stage effectively translates the physical movements into meaningful, grammatically correct textual sentences, reflecting the intent of the signer.

The final step involves converting the generated text into verbal speech using Kokoro AI.Kokoro AI uses advanced speech synthesis techniques to transform the text into natural, human-like speech.It captures the nuances of tone, pitch, and emphasis, providing a more human-like conversational experience.

CHAPTER 4

COMPONENTS

4.1 SOFTWARE COMPONENTS

4.1.1 LLAVA-NLP

LLAVA-NLP (LLaVA - Language and Vision Assistant) is a state-of- the-art, multimodal AI system that combines language processing with visual comprehension. It takes the building blocks of large language models (LLMs) such as OpenAI's GPT or Meta's LLaMA and expands their functionality to process both text and visual information in parallel. LLAVA-NLP is trained to understand, generate, and converse using text and images and is thus particularly well-suited to applications that demand a high level of comprehension of both linguistic and visual context. This enables the model to handle sophisticated multimodal inputs, and it is thus applicable in domains such as image captioning, visual question answering, medical image analysis, and real-world robotics.

LLAVA-NLP works by merging the capabilities of vision transformers (ViTs) for image processing with sophisticated language models for text generation. It receives an input image and corresponding text, processes the visual information through a vision encoder to get meaningful features, and then feeds these features into a language model for context-based text generation. The fusion is usually realized through cross-modal attention mechanisms, which aligns visual and textual features to generate coherent, contextually correct outputs. This enables LLAVA-NLP to comprehend and narrate images, respond to questions grounded in visual context, and even produce dialogues that are visually aware.

One of the most notable aspects of LLaVA is its contextual understanding, which is attained through cross-modal attention mechanisms. These mechanisms enable the model to map visual features to textual context, improving its capacity to produce accurate and contextually appropriate responses. Moreover, LLaVA utilizes pre-trained large language models, guaranteeing high accuracy in language generation and comprehension.

The scalability of the model is another major benefit. LLaVA can be adapted to specialized applications, such as medical diagnosis, self-driving cars, and customer support, and hence it is versatile for different domains. Its applicability is not only limited to academic domains but also to real-world applications, including video analysis, AR/VR, human-computer interaction, and assistive technologies, illustrating its wide potential across different fields.

LLaVA has some advantages that make it more useful in a variety of applications. Its multimodal aspect can process text and image inputs, making it possible to understand content at a deeper level. Visual instruction tuning allows LLaVA to execute complex visual instruction-based tasks, enhancing its performance in image captioning and visual question answering. The high accuracy of the model is obtained by using pre-trained large language models, which provide precise language generation and comprehension.

The scalability of LLaVA enables it to be fine-tuned for particular use cases, including medical diagnosis, self-driving cars, and customer support, and hence can be used in a variety of domains. Its applications in real-world scenarios range from video analysis, augmented and virtual reality, human-computer interaction, and assistive technologies to showing its versatility and potential influence across industries.

4.1.2 KOKORO-AI

Kokoro-AI, also known as Kokoro TTS, is a cutting-edge, open-source text-to-speech (TTS) model designed to convert written text into natural, human-like speech. Built on the efficient StyleTTS2 architecture, this model strikes a remarkable balance between high-quality output and computational efficiency, boasting just 82 million parameters. Despite its compact size, Kokoro-AI produces impressive audio quality comparable to larger, more resource-intensive models, making it an attractive choice for both developers and businesses.

Kokoro-AI's operation is streamlined and effective, involving three main stages. First, it processes input text, breaking it down into tokens and extracting linguistic features to understand the structure and meaning of the text. Next, it leverages the StyleTTS2 framework to generate intermediate acoustic representations, capturing nuances like prosody, intonation, and emotion. Finally, the model employs the ISTFTNet vocoder to transform these acoustic features into actual waveform audio, resulting in clear, realistic speech. This layered approach allows Kokoro-AI to deliver natural, expressive voices with low latency, making it ideal for real-time applications like virtual assistants, call center automation, and interactive voice response (IVR) systems.

Several factors make Kokoro-AI a standout choice for TTS applications. Its relatively small size reduces computational demands, enabling faster processing and more cost-effective deployment. Additionally, being open-source and licensed under Apache 2.0, it offers significant flexibility for developers to modify, distribute, and commercialize their projects without licensing restrictions.

Kokoro-AI also supports multiple languages, including American and British English, French, Korean, Japanese, and Mandarin, with customizable voice packs for different tones and styles. This flexibility allows businesses to tailor speech outputs to specific audiences, enhancing customer experiences and engagement.

The need for Kokoro-AI stems from the growing demand for efficient, real-time speech synthesis across industries. Unlike traditional, bulky TTS models, Kokoro-AI provides a powerful alternative for devices with limited resources, such as IoT devices, smart speakers, and mobile apps. It also addresses the increasing focus on personalization in voice technology, allowing companies to create custom voices that align closely with their brand identity. As an open-source model, it further democratizes access to high-quality TTS technology, fostering innovation and reducing the barrier to entry for smaller companies and independent developers.

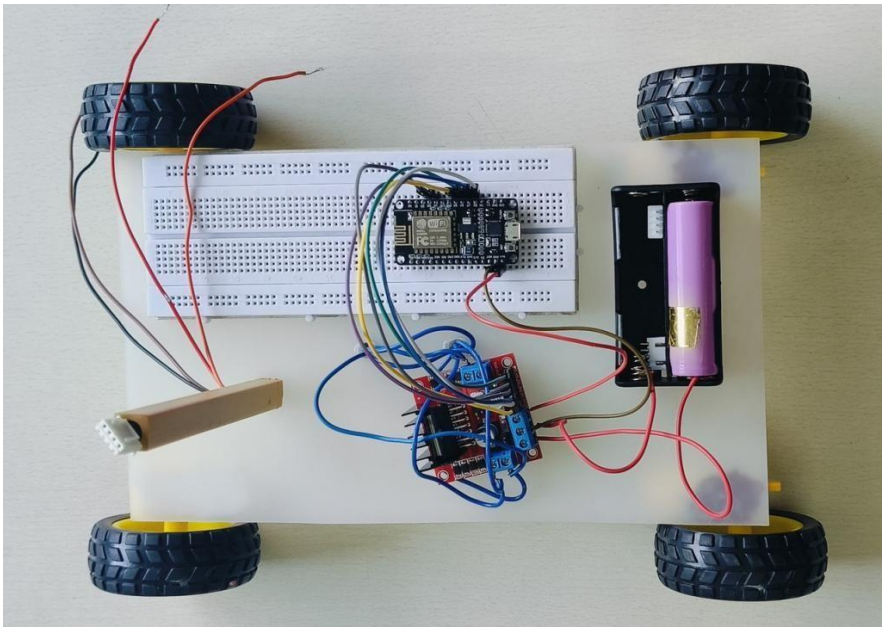
To get started with Kokoro-AI, users can access the model through its Hugging Face repository, install necessary dependencies, and choose from various available voice packs to match their application's tone and style. This streamlined setup makes it an excellent choice for anyone looking to integrate speech capabilities into their software or services.

In summary, Kokoro-AI stands out as a highly efficient, versatile, and accessible text-to-speech solution, making it a compelling option for a wide range of applications, from virtual assistants to multilingual customer support systems. Its blend of performance, flexibility, and open-source availability positions it as a key player in the evolving field of voice technology.

CHAPTER 5

RESULT AND DISCUSSION

In conclusion, "Bridging silence: real time sign-language translator using webcam" our proposed project powered by LLaVA-NLP and Kokoro TTS, represents a significant step toward breaking communication barriers for mute individuals. By offering immediate, accurate, and expressive translations without the need for human interpreters, this system promotes greater self-reliance and inclusion. Its streamlined design, utilizing standard webcams for real-time processing, ensures it remains accessible and practical in diverse settings. Ultimately, this innovation holds the potential to transform the way mute individuals engage with the world, fostering a more connected and inclusive society.



CHAPTER 6

REFERNCES

REFERENCES

[1]. Abdulmalek, S., Nasir, A., Jabbar, W. A., Almuhaya, M. A., Bairagi, A. K., Khan, M. A. M., & Kee, S. H. (2022, October). IoT-based healthcare-monitoring system towards improving quality of life: A review. In Healthcare (Vol. 10, No. 10, p. 1993). MDPI.

[2]. Junaid, S. B., Imam, A. A., Balogun, A. O., De Silva, L. C., Surakat, Y. A., Kumar, G., ... & Mahamad, S. (2022, October). Recent advancements in emerging technologies for healthcare management systems: a survey. In Healthcare (Vol. 10, No. 10, p. 1940). MDPI.

[3]. Ramesh, V., Sankaramahalingam, M., Bharathy, M. D., & Aksha, R. (2017, July). Remote temperature monitoring and control using IoT. In 2017 International Conference on Computing Methodologies and Communication (ICCMC) (pp. 1059-1063). IEEE.

[4]. Gada, U., Joshi, B., Kadam, S., Jain, N., Kodeboyina, S., & Menon, R. (2021, January). Iot based temperature monitoring system. In 2021 4th Biennial international conference on Nascent technologies in engineering (ICNTE) (pp. 1-6). IEEE.

[5]. Rao, V., & Prema, K. V. (2018, May). Internet-of-things based smart temperature monitoring system. In 2018 3rd IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT) (pp. 72-77). IEEE.

[6]. Tamura, T., Zhou, J., Mizukami, H., & Togawa, T. (1993). A system for monitoring temperature in bed and its application to the assessment of body movement. Physiological measurement, 14(1), 33.

[7]. Gulcharan, N. F. B. I., Daud, H., Nor, N. M., Ibrahim, T., & Shamsudin, M. Z. (2014). Investigation of stability and reliability of the patient's wireless temperature monitoring device. Procedia Computer Science, 40, 151-159.